

Planning Shared Review for Language Model Workflows with Expiring Intervention Opportunities

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Abstract: A shared reviewer must decide which language model proposals can benefit from intervention before their review opportunities expire. We formulate allocation with finite service time and application-specific expected preventable loss, including corrections and harmful reviews. Our principal case study uses measured Berkeley Lab air-handler observations with experimentally imposed fault labels. A frozen evaluation covers 18 whole-day blocks on one laboratory unit. The model diagnoses 19 of 54 windows correctly, versus 27 for a conventional classifier. Model review corrects one error but makes six correct proposals unresolved. A development-fitted signed-benefit rule declines review, producing a mechanical policy tie. With ideal review, search reduces modeled loss relative to greedy by 3.333 points, but earliest-deadline-first matches search. Controlled synthetic maintenance provides complementary evidence of conditional planning benefits and objective tradeoffs. Adapted retail workflows retain unfavorable search-minus-greedy results of +0.250 and +1.667 under two intervention authorities. Together, the studies show how capacity, risk ranking and corrective effectiveness determine whether review ordering matters, while separating externally validated task correctness from assumed oversight dynamics.

1 INTRODUCTION

Several agents may each have a reasonable request for assistance, yet one reviewer cannot address them all before their actions take effect. Reviewing the request with the largest immediate expected benefit may consume the only remaining slot for another useful intervention. Conversely, planning an elaborate review order may accomplish no more than earliest-deadline-first when capacity is sufficient, or may protect a correct transaction while a later mistake expires.¹

These possibilities arise in workflows where intervention has a service time and a boundary beyond which it cannot undo a consequence. A maintenance correction can stop future downtime without erasing downtime already incurred. A retail approval gate can prevent a mistaken transaction from being posted, but may lack authority to replace it with the intended transaction. The value of a review therefore depends on when it finishes and what the reviewer is allowed to do.

We ask when planning the order of a finite shared review queue improves outcomes over greedy benefit allocation and earliest-deadline-first. We also examine how risk estimates, capacity and the performance

objective qualify any benefit. Our principal application uses measured laboratory HVAC observations and saved model diagnoses and reviews. A synthetic maintenance environment isolates scheduling mechanisms, while executable retail workflows test transfer and intervention limits.

The contributions are threefold. First, we give an executable formulation that connects public review requests to an application-specific preventable-loss function, while keeping hidden correctness labels outside scheduling. Second, we implement complementary protocols for feedback-dependent maintenance trajectories and allocation on identical saved HVAC judgments or retail transactions. Third, we provide externally grounded diagnosis evidence alongside frozen paired planning gains, null results and unfavorable transfer results. The measured study makes reviewer effectiveness a tested component rather than an exclusively perfect assumption. Exact enumeration supplies an interpretable small-queue planner. The contribution concerns the formulation, integration and empirical analysis rather than a new exhaustive scheduling algorithm.

¹Drafting assistance: OpenAI Codex (OpenAI, 2026). See the AI assistance disclosure.

2 RELATED WORK

Help-seeking, communication and constrained deferral address related parts of the oversight problem. KnowNo uses calibrated prediction sets to decide when a language model planner should request help (Ren et al., 2023). Its request-admission mechanism and coverage assumptions differ from ordering an already admitted queue. Our uncertainty-first baseline ranks estimated error probability; it does not reproduce KnowNo or provide a conformal guarantee.²

Value of Information evaluates clarification by its expected decision utility relative to communication cost (Dong et al., 2026). We similarly express intervention value through an explicit objective, but allocate a reviewer among competing pending transactions rather than choose a clarification question. One Human, N Agents studies a limited audit budget under miscalibrated and correlated confidence (Zavattari et al., 2026). That work directly establishes shared auditing as prior research. Our focus is finite-duration intervention before a posting or correction cutoff, including the difference between blocking an error and completing a task.

DeCCaF learns expert correctness and uses cost-sensitive constrained allocation across experts (Alves et al., 2024). We retain an ideal-review reference and separately measure corrections and harm from a model reviewer on sensor evidence. These are conceptual comparisons of mechanisms, not reproductions of those complete systems. Author-linked implementations of KnowNo, Value of Information and DeCCaF were inspected. No author-linked released implementation for One Human, N Agents was located in the checked paper and project searches.

Classical scheduling already studies deadlines, processing time and weighted consequences. Hariri et al. consider weighted late work, meaning processing performed after a due date (Hariri et al., 1995). Guo et al. analyze a Pareto tradeoff between weighted tardy-job counts and weighted late work (Guo et al., 2022). Our losses differ, but the connection explains why urgency, count objectives and cost objectives can favor different schedules. Exhaustive current-queue search is an established method for a bounded scheduling problem.

The retail substrate is τ -bench, which supplies interactive tasks, domain policies, database tools and annotated final-state validation (Yao et al., 2024). We reuse its pinned retail implementation while changing the user interaction and adding staging, cutoffs and review. This adaptation makes transaction review

testable while retaining genuine retrieval, item selection, payment and mutation semantics.

3 REVIEW ALLOCATION MODEL

A released review request i exposes an arrival time a_i , a cutoff d_i , a proposed action, review duration s_i , public consequence parameters and an estimated initial-error probability p_i . The scheduler sees only released requests. It cannot inspect a hidden correct action, a future job or a future model output. Each nonpreemptive reviewer processes one request at a time. The measured HVAC application also varies the number of reviewer slots. A review result becomes available only when its service finishes.³

Let $G_i(T)$ be the preventable consequence if request i is initially wrong and an effective review completes at time T . The expected benefit is

$$B_i(T) = p_i G_i(T). \quad (1)$$

Eligibility requires release and positive benefit from starting now, $B_i(t + s_i) > 0$. A cutoff equality is timely. Consequences incurred before completion remain incurred. This expression assumes effective review. The HVAC study replaces error risk with signed expected improvement to include ineffective and harmful model reviews. Ideal correction remains a reference, and retail separately varies corrective authority.

3.1 Policies and Planning

FCFS selects the earliest eligible arrival. Earliest-deadline-first (EDF) selects the earliest cutoff. Uncertainty-first selects the largest p_i . Delay-aware greedy selects the largest $B_i(t + s_i)$. Queue-order search enumerates ordered subsets π of the eligible current queue and maximizes

$$\sum_{k=1}^{|\pi|} B_{\pi_k} \left(t + \sum_{j=1}^k s_{\pi_j} \right). \quad (2)$$

It dispatches the first request and replans when free. It does not plan with unreleased arrivals or hidden labels, and it is not a full-horizon optimal-control solution. Ties prefer fewer reviews and then the declared lexicographic arrival, agent and request order. Value sums use stable summation. Historical numerical audits distinguish exactly co-optimal tie differences from positive decision regret.

²Drafting assistance: OpenAI Codex (OpenAI, 2026). See the AI assistance disclosure.

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There are $\sum_{k=0}^q q!/(q-k)!$ ordered subsets for q eligible requests. This is 16 at three requests and 1,957 at six. We enumerate them all without queue truncation. Retail additionally includes a no-review control with the same automatic safeguards. Maintenance includes a myopic-benefit baseline; its terminal-only greedy equivalence is checked rather than presented as independent evidence.

3.2 Pairing and Interpretation

An exogenous scenario fixes jobs or source cases, arrivals, deadlines and costs. A policy run is not an independent scenario. Maintenance conditions share exogenous streams but receive fresh generations and separate feedback histories. Retail conditions share a saved prepared transaction within a generation replicate because no model generation follows staging. HVAC conditions similarly share prepared diagnoses and review responses. These comparisons isolate allocation on identical inputs, while synthetic maintenance measures integrated trajectories that may also differ in model behavior.

For each declared primary comparison, we compute scenario-level paired loss differences and a percentile interval from 2,000 bootstrap resamples. Retail first averages three generation replicates within each source bundle. All conditions for a scenario are resampled together. Other contrasts are secondary, with no multiplicity adjustment. A wide interval spanning zero is uncertainty about the effect, not evidence of equivalence.

4 MEASURED HVAC DIAGNOSIS STUDY

Our principal externally grounded application is review of diagnoses from measured air-handler observations. Berkeley Lab’s public collection includes both physical experiments and simulations (Granderson et al., 2020). We use only its SZCAV and SZVAV physical FLEXLAB records, pinned to Figshare version 3. The experiments imposed component faults on the air handler serving test cell X3A. They are laboratory measurements with controlled faults, not synthetic sensor observations or naturally occurring faults in occupied buildings.⁴

⁴Drafting assistance: OpenAI Codex (OpenAI, 2026). See the AI assistance disclosure.

4.1 Evidence, Labels and Source Split

The selected records contain 26 experimental days under constant-volume and variable-volume control. The inventory supplies the imposed condition for each day. We group stuck and leaking variants into four supported categories: normal, outdoor damper, heating valve and cooling valve. The binary fault column is cross-checked offline. Dates, filenames, fault indicators and descriptions revealing the imposed condition are withheld from model and scheduler inputs.

Each day provides three complete 60-minute windows beginning at 10:00, 12:00 and 14:00. Ten channels describe supply, outdoor, mixed and return temperatures, supply setpoints, and fan, damper and valve commands. The model receives explicitly named means, minima, maxima and changes from first to last reading. Fahrenheit units, fractional commands and recording hours are retained. Commands are not assumed to measure actual component position. No later observations enter a window’s evidence. Figure 1 connects a recorded example to its generated diagnosis.

The first chronological day of each mode and category is development, giving eight days and 24 windows. The remaining 18 days and 54 windows form the frozen evaluation. Adjacent windows never cross the split. All included days are selected from source records before evaluation generation, without filtering on errors or policy performance. A standardized nearest-centroid classifier fits its normalization and class centers using development data alone and uses the same statistics, control mode and hour as the model.

4.2 Fallible Review and Expected Benefit

The pinned Qwen model generates one diagnosis and then a separately prompted review of that diagnosis and the same permitted observations. The review may preserve the category, change it or abstain. These responses are correlated same-model judgments, not independent experts. An initial development pilot exposed ambiguous numerical arrays and frequent abstention. One declared revision names every statistic and requests the most likely supported category. All failed pilot outcomes remain saved. No further prompt change follows evaluation.

For an error indicator E_i before review and E_i^R after review, define the conditional expected accuracy gain

$$g_i = \Pr(E_i = 1, E_i^R = 0 \mid x_i) - \Pr(E_i = 0, E_i^R = 1 \mid x_i). \quad (3)$$

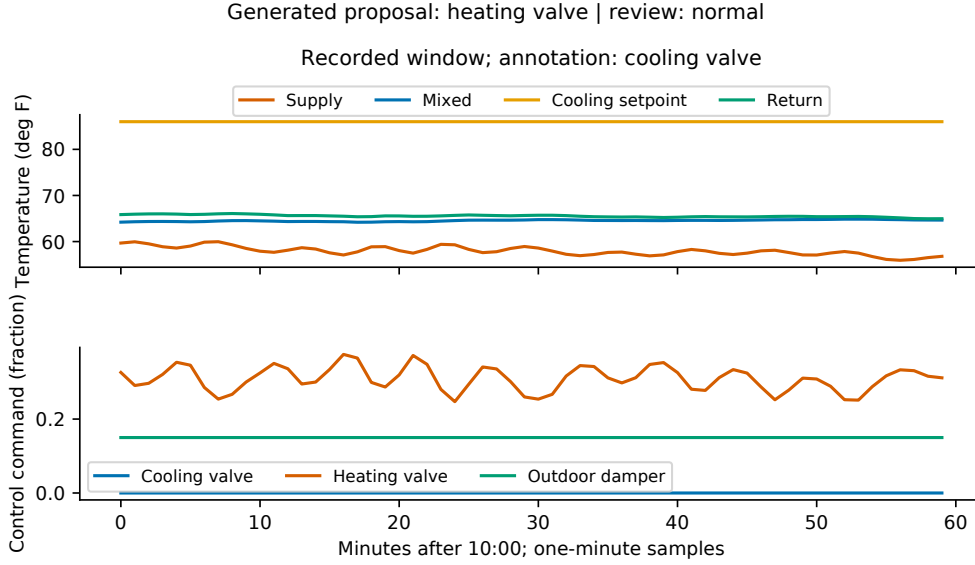


Figure 1: Measured temperatures and control commands from the first source-order evaluation window. The cooling-valve annotation comes from the experiment inventory and is shown only for analysis. The agent receives window summaries, without that annotation or source date. Both its generated heating-valve proposal and the model review’s normal diagnosis are wrong. Recorded cooling despite a zero cooling-valve command illustrates information that the weak proposer failed to use reliably.

Here x_i is public proposal information. A harmful abstention counts as an unresolved task. The modeled benefit of timely review is $w_i g_i$, rather than $w_i p_i$. We estimate risk by $(e + 1)/(n + 2)$ and signed gain by $(c - h)/(n + 2)$ from development errors e , corrections c and harms h . Proposed-category bins with fewer than six examples use the pooled estimate. This is a transparent small-data shrinkage rule, not a well-established reviewer model.

The revised development set has six correct proposals, one correction and one harmful review. Pooled gain is zero, and the populated heating-proposal bin has negative gain. Thus positive-benefit eligibility declines every review in the primary condition. This mathematical consequence is frozen before evaluation. A secondary risk-only condition deliberately ignores reviewer harm while applying the same saved responses. An ideal condition returns the annotated category only when selected review completes. It bounds reviewer effectiveness, without giving hidden labels to scheduling.

4.3 Constructed Oversight Dynamics

The three windows become retrospective diagnosis tickets after 14:59, and all are released at abstract tick zero. They describe the same equipment and imposed

condition, not three independent machines. The experiment models allocation across competing diagnostic outputs, not physical repair. Cutoffs 2, 4 and 6 and dimensionless error weights 4, 8 and 12 are independently permuted by a public run seed. An incorrect or unresolved ticket incurs its weight at cutoff. A timely completed review can change the diagnosis before that point. It cannot undo a past consequence or establish subsequent repair success.

We freeze one or two reviewer slots, service durations of one, two or three ticks, and a uniform-weight sensitivity with weight eight. Equal-time completion is timely. Search enumerates the current queue and assigns proposed reviews to the next available slot. Measured GPU latency is recorded separately from these assumed service times. All six policies reuse identical saved proposals and review responses. This yields 3,888 paired policy replays without additional generations.

The primary contrast is search minus greedy loss with signed-gain model review, one slot, two-tick service and heterogeneous weights. We resample whole experimental days 2,000 times with seed 20260917, stratifying by control mode and keeping all matched conditions together. Eighteen day blocks on one piece of equipment support conditional within-facility summaries. With one physical unit there is no stable equipment-level interval. Day-block exchangeability

is unverified, so the intervals are conditional resampling summaries. The 54 windows are not 54 independent cases.

4.4 Diagnostic and Allocation Results

The proposer correctly labels 19 of 54 windows (35.2%), versus 27 (50.0%) for the conventional classifier. It proposes a heating-valve fault on 51 windows, correctly identifying all 18 heating cases but none of the 15 cooling or 12 damper cases. The conventional baseline identifies 14 heating, seven damper and six normal windows, but also misses every cooling case. The task remains difficult for both methods. Day-level imposed-condition labels do not guarantee that each selected window excites a detectable fault symptom. Low accuracy therefore reflects the full representation and diagnosis task, not an isolated measure of reasoning skill. These results establish a diagnosis bottleneck rather than a reliable deployed detector.

Across saved review pairs, review corrects one of 35 wrong proposals and leaves 34 unresolved. It preserves 13 of 19 correct proposals, changes five to a wrong category and abstains on one. Five review responses abstain in total. There are no malformed proposals, malformed reviews or failed generation attempts in the final evaluation. Review-everything accuracy is 14/54, below initial accuracy. Frozen risk has Brier score 0.2305 and AUROC 0.5023. Sparse, nearly tied risks provide little discrimination.

All primary policies decline review and have mean loss 15.333 with 19 correct diagnoses. Search minus greedy is exactly zero on all 18 day blocks. The degenerate paired interval describes a shared frozen admission rule, not evidence that scheduling policies are equivalent. Ignoring reviewer harm produces search and greedy loss 18.000, although search performs 54 reviews versus 39. Their sequences differ on 15 days with no loss difference. Risk-only search is worse than no review by 2.667 points, with interval [0.217, 5.778].

Under ideal review, search attains zero loss versus greedy’s 3.333, a paired difference of -3.333 [$-4.889, -1.778$], with ten day wins, eight ties and no losses. EDF also attains zero and chooses the same sequence as search. With three simultaneous equal-duration requests and cutoffs 2, 4 and 6, earliest-deadline ordering can fit every two-tick review. This structural explanation limits the interpretation of the ideal-reference advantage. Figure 2 shows correctness alongside modeled loss. Capacity sensitivity, all day differences and first-qualifying timelines are in the supplement. No measured equipment damage,

energy saving or human-review effectiveness follows from these results.

5 SYNTHETIC MECHANISM STUDY

A separate civilian maintenance simulation exposes binary faults through invented noisy clues with accuracies 0.8 and 0.6. It is not measured ventilation data. Two fresh model samples accompany each job; the first executes and the second supplies agreement information. Completed ideal supervision can correct an action and change later feedback histories. An error accumulates downtime until correction or closure, plus a terminal penalty. Public arrivals, costs and deadlines are independent of hidden faults.⁵

The frozen evaluation contains 2,752 episodes across original, constructed competition, larger workloads and an objective extension. The supplement preserves complete distributions, risk definitions, matrices and results. Search minus greedy at two-tick service is -0.094 [$-0.563, 0.344$] on 64 original scenarios and -1.313 [$-2.250, -0.438$] on 64 competition scenarios. Competition releases three requests together and uses terminal penalties only. Search beats EDF there by 2.563 points at two ticks, while both reach zero loss at one tick.

On the six-agent workload, search minus greedy is -6.188 [$-9.376, -3.281$] at one tick and -0.219 [$-2.313, 1.969$] at two ticks. Lower capacity does not monotonically increase the value of planning. Analytical risk improves prediction scores without consistent allocation gains. Adding an incorrect-closure penalty reduces wrong closures at one tick while increasing original cost. These positive, null and unfavorable findings illustrate mechanisms and objective tradeoffs without providing external task validation.

6 RETAIL TRANSFER AND INTERVENTION LIMITS

The retail application adapts pinned τ -bench tools, policies and task records (Yao et al., 2024). Customer and transaction records are simulated. GPU-generated agents retrieve accounts, interpret policies, confirm intent and propose validated cancellations, modifications or returns/exchanges. Automatic guards remain enabled. Completed simulated review may intervene at an added staging boundary before a

⁵Drafting assistance: OpenAI Codex (OpenAI, 2026). See the AI assistance disclosure.

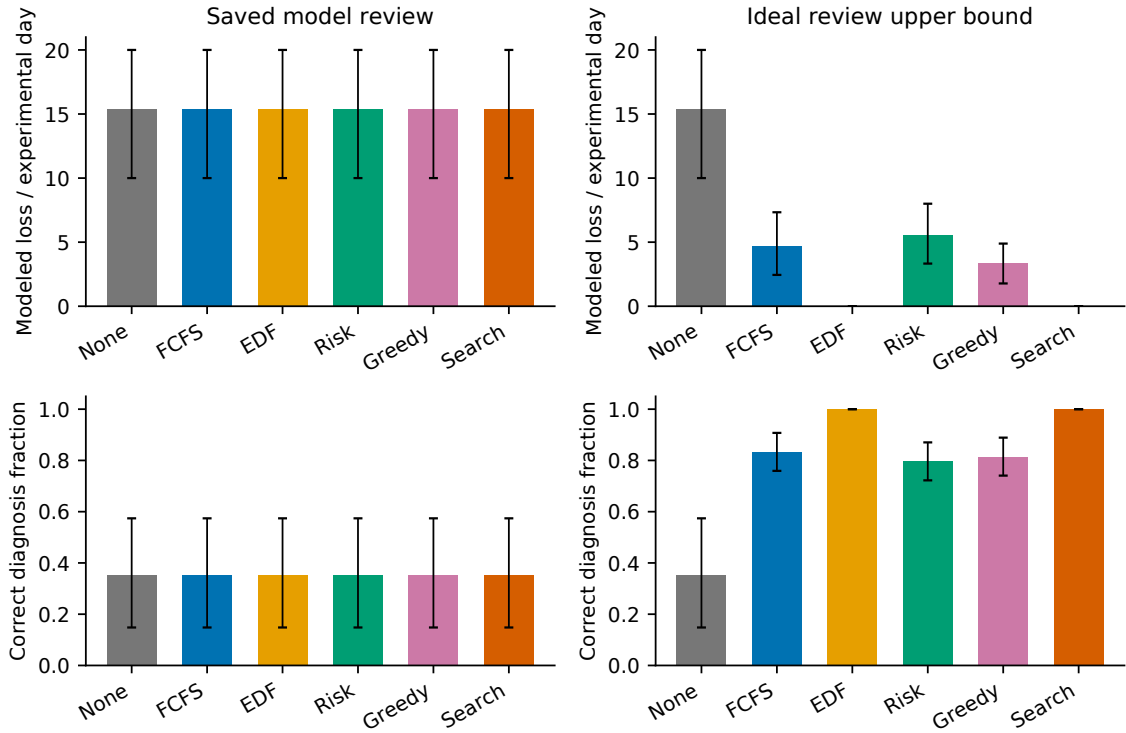


Figure 2: Externally annotated diagnosis correctness and simulated weighted loss, one reviewer and two-tick service. The left condition uses development-fitted signed model-review benefit and admits no reviews. The right assumes ideal completed review. Proposals are generated from measured windows; loss weights and scheduling dynamics are constructed. Bars show 95% day-block bootstrap intervals conditional on one facility, with all paired windows and conditions retained.

constructed cutoff. This adaptation does not produce an official benchmark score.⁶

The operational objective combines an unresolved-service cost of four with a dimensionless wrong-posting weight. Original ideal review can replace a staged transaction with its annotated target. A restricted reviewer can instead approve or block, preventing a wrong posting while leaving the service request unresolved. A separate complexity condition adds one tick for multi-item proposals. Customer reconfirmation after replacement, real processing cutoffs and empirical consequence weights remain unresolved assumptions in the source-based methodological review.

The original evaluation uses 96 source cases in 32 bundles, three generation replicates and 288 GPU workflows. All 1,152 policy replays share preparations within each replicate. Fifty-five unstaged failures remain outside the review boundary. At one tick every review policy corrects every staged error. At two ticks, search minus greedy mean loss is +0.250 with a 95% bundle-paired interval $[-0.333, 1.000]$.

⁶Drafting assistance: OpenAI Codex (OpenAI, 2026). See the AI assistance disclosure.

one bundle win, 29 ties and two losses. Search completes 222 jobs versus greedy’s 223 and has mean loss 3.708 versus 3.458. This primary result is unfavorable to search.

The robustness follow-up excludes all 120 previously used accounts across upstream splits. It evaluates 24 fresh cases in eight bundles, again with three replicates. Under restricted authority and two-tick service, search minus greedy is +1.667 $[0.000, 4.667]$, with zero wins, six ties and two losses. Search blocks eight wrong transactions versus greedy’s twelve, while both complete the same 39 initially correct tasks. Twenty-one unstaged failures remain. Full correction and complexity time both give a +2.333 search-minus-greedy difference, although complexity raises both losses. Post hoc reuse of the original preparations is reported separately in the supplement.

These negative results have explanatory value. Preserving a public review opportunity can prioritize a correct transaction while a wrong one expires. Restricted authority changes which consequence is preventable without necessarily changing review order. More completed reviews do not imply more corrected

tasks. The main claims therefore retain both unfavorable retail comparisons rather than treating synthetic planning gains as a universal result.

7 DISCUSSION

Planning helps when interventions are effective, compete for time and ordering preserves opportunities that greedy selection would lose. The measured application shows why effectiveness must be checked first. A frozen signed-gain estimate declines harmful model review; ideal review creates an ordering benefit that EDF already captures in the simple deadline structure. Constructed maintenance competition provides that condition. The original maintenance workload and the retail study show why a favorable mechanism does not imply a general policy ranking. Public contention can involve correct requests, some errors are already infeasible, and later arrivals are outside the planner’s current state.⁷

Capacity changes the opportunity set. When every useful queued error can be corrected, simple review policies can match search despite different orders. When capacity is severely constrained, too few feasible combinations may remain for ordering to help. More completed reviews can then coexist with fewer corrections. The meaningful outcome is preventable operational consequence, not sequence diversity alone.

The reviewer objective and authority must also be explicit. A search policy that optimizes weighted cost can close fewer jobs correctly. A block-only reviewer can prevent processing harm while leaving service unfinished. These outcomes are consistent with their objectives and should not be converted into claims of universal superiority. The robustness study isolates these mechanisms without changing risk estimates or agent preparation.

Our evidence is limited to one model and bounded workloads. Measured HVAC experiments cover one laboratory unit, two control modes and deliberately imposed faults. Their 18 evaluation days are correlated through common equipment, and both diagnosis methods miss the cooling class. Model review is weak and correlated with proposal errors. Synthetic faults supply a transparent mechanism study, while retail records offer executable tools but shared templates and possible training exposure. Scripted confirmation and complete-transaction correction simplify customer interaction. Deadlines, processing weights

and complexity times are constructed quantities. Exact annotated equality can overstate some operational distinctions. No measured human review performance, retailer savings or large-queue scalability follows.

8 CONCLUSION

Shared review is a timed intervention problem with an application-specific preventable-loss function. Measured HVAC evidence establishes reviewer effectiveness as a central bottleneck. A development-fitted signed-benefit rule declines model reviews that often introduce errors, while ideal review creates an ordering benefit already captured by EDF. Synthetic maintenance supplies complementary planning gains and objective tradeoffs. Both retail primary comparisons remain unfavorable to search. Together, these results support analyzing capacity, risk ranking and corrective authority alongside scheduling. External diagnosis labels establish task correctness, while operational consequences remain assumptions. The framework makes this separation reproducible without claiming human performance or real savings.⁸

AI ASSISTANCE AND REPRODUCIBILITY

OpenAI Codex assisted implementation, test and analysis code, source checking, figure code, and drafting and revision throughout this manuscript and supplement (OpenAI, 2026). Numerical results derive from measured source windows, annotated conditions, logged GPU generations and deterministic simulation, with declared tests and replay checks. Figures plot those saved measurements. AI is not an author. Human authors remain responsible for verification, interpretation and submission. Source snapshots, declarations, raw evidence, ledgers and reproduction commands accompany the local research package. Previously disclosed publication redactions remove unsolicited credential-like model text from auxiliary arguments and prompt histories, preserving original hashes and unchanged transactions, labels and outcomes. Exact originals remain on the authorized pod. Deterministic replay does not promise bitwise regeneration from a seed.

⁷Drafting assistance: OpenAI Codex (OpenAI, 2026). See the AI assistance disclosure.

⁸Drafting assistance: OpenAI Codex (OpenAI, 2026). See the AI assistance disclosure.

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